

Senior Thesis

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```
rhfs <- read.csv("RHFS_2024_Public_Variables.csv")

# Turning Categorical Variables into Factors
rhfs <- rhfs |>
  mutate(
    across(
      c(NUMCAT_R, OUTCOME, PROPCON, PROPTOWN,
        COMPLEX, NUMBLD_R, YRNEWBLG_R, YRPROPACQ_R, APPRWAIVER,
        CASHDWN, HOWACQ, HOWMVAL, OFFSTPARKAHS, UNITSTAT,
        ADAGENCY, ADAGENT, ADNEWSPAPER, ADONLINE, ADOTHONLINE,
        ADVERTISE, ADWEBSITE, MLS, MNGMNT, NUMOTHPROP,
        NUMOTHUNITS, OWNENT, OWNOTHPROP, COMMSPACE, CONCSS,
        ICABLE, IELEC, IGAS, IPARKING, IPOOL,
        ISEWER, ITRASH, IWATER, IWIFI, LEASLNG,
        LNGTRMPRV, NORNTAPPFEE, PAYAPP, PAYCASH, PAYCHK,
        PAYCREDIT, PAYONLINE, PAYOTH, PAYTHIRD, RCONTROLAHS,
        RECOMMREN, RNTAPPFEWAIV, SECDEPAMT, SENIOR, WEEKLNG,
        OPTAX_R, PROPANS, ACCEPT, BENEOTH, ECONDEVL,
        ELDERLY, GOVBELOW, GOVGRANT, HISTORIC, LOWINAREA,
        LOWINTEN, OTHRCV, RSUBSIDY, STLOWINC, YRMORT_R,
        DEBT, DEBT1, GREENMORT, LOCTYP, MORTGET,
        PMTFREQ, REGPAY, RPIPMI, RPIPRIN, RPIPRPINS,
        RPIPRPTAX, WHYNOMRT, SAMPSTRATA, MORTAMT_R, OUTBAL_R, RNTAPPFEE_R,
        YRPROPREHAB_R),
      as.factor
    )
  )

# Loads the data frame with srvyr
df <- rhfs |>
  select(-REPWGTO) |> # REPWGTO is equivalent to WEIGHT
  as_survey_rep(
    repweights = starts_with("REPWGT"),
    weights = WEIGHT,
    type = "successive-difference",
    combined_weights = TRUE,
    # Decided to set to T to follow SDR as outlined by Ash (to match what was done in Census's Methodol
    mse = TRUE
  )

# Sanity check for if the data was successfully loaded with srvyr
# using Table 5 of Census's RHFS Methodology
```

```
df |>
  summarize(
    properties = survey_total(vartype = "se") # Applies weights and calculates se of properties
  ) |>
  mutate( # Dividing by 1000 to match Table 5's scale
    properties = properties / 1000,
    properties_se = properties_se / 1000,
    moe = properties_se * 1.645 # 90% margin of error
  )
```

```
## # A tibble: 1 x 3
##   properties properties_se   moe
##   <dbl>         <dbl> <dbl>
## 1    18965.         165.  271.
```

```
# Makes View(rhfs) much more productive.
rhfs <- rhfs |>
  select(-starts_with("REPWGT"), -WEIGHT,
    -SURVYR, -CONTROLPUF, # is always 2024, ID isn't useful in data analysis
    # J flag variables mark values that were manually reviewed; note that
    # they may become relevant if outliers are under scrutiny.
    -starts_with("J"),
    # A continuous variable coded as a factor without documentation; may
    # be useful for pure prediction
    -PCTFEWAIV_R
  )

# Renaming some factor levels for later convenience
rhfs <- rhfs |>
  mutate(
    NUMCAT_R = fct_recode(
      NUMCAT_R,
      "1 Unit" = "0",
      "2-4 Units" = "1",
      "5-24 Units" = "2",
      "25-49 Units" = "3",
      "50+ Units" = "4"
    ),
    OUTCOME = fct_recode(
      OUTCOME,
      "Interview Compl; self response" = "1",
      "Assisted living limited interview; self response" = "3",
      "Conversion from partial to full interview" = "5",
      "Interview Compl; field interview" = "201",
      "Assisted living limited interview; field interview" = "203",
      "Sufficient partial; field interview" = "204",
      "Military limited field interview" = "206"
    )
  )
```

```
# No explicit NAs
sum(is.na(df))
```

[1] 0